

Prakrit Ramachandran

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EDUCATION

University of Washington — B.S. Mechanical Engineering

Coursework: Statics, Kinematics, Mechanics of Materials, Materials Science, Thermodynamics

Seattle, WA

2024 - Present

Mission San Jose High School (UW GPA: 3.9)

Activities: 3D Printing Club | Tech & Robotics Club | DECA

Fremont, CA

2020 - 2024

EXPERIENCE

Mechanical Engineer

Seattle, WA

Husky Robotics - Drone Subteam

Jan 2025 - Present

Led mechanical design and fabrication of a competition drone within a 10-week build cycle; delivered a flight-ready system meeting payload and structural constraints

Designed and validated a custom multi-orientation gripping mechanism to improve object capture reliability during dynamic flight

Produced detailed CAD assemblies and manufacturing drawings; performed material selection and structural simulations to support design reviews

Team Captain

Fremont, CA

FIRST Tech Challenge

Aug 2018 - Mar 2024

Recruited and led an 11-member engineering team while managing three functional subteams and a \$10K annual budget

Directed mechanical architecture, detailed design, and assembly of six 500+ part competition robots under strict seasonal deadlines

Integrated mechanical, electrical, and software subsystems to enable autonomous navigation and precision object manipulation

Competed at Regional, World Championship, and Chicago Invitational levels; earned 30+ team awards

Branch Captain

Fremont, CA

FIRST Robotic Competition

Jan 2022 - Mar 2024

Directed mechanical design initiatives for a 10-member subteam within a 40+ member organization during 6-week build seasons

Designed two 800+ part robotic platforms integrating drivetrain, manipulator, and structural subsystems

Manufactured 50+ precision components using CNC machining and supported full system assembly, testing, and refinement

Facilities Engineer Intern

Seattle, WA

Quanta Labs

Jun 2023 - Aug 2023

Collaborated with Facilities and Quality teams to develop detailed mechanical schematics and standardized documentation

Designed four AutoCAD schematics for shock testing equipment, supporting manufacturing and deployment timelines

Contributed to PCB layout development for vibration testing systems to improve measurement accuracy and system reliability

PROJECTS

Marble Coaster

Designed & built a fully custom continuous-motion marble coaster, progressing from concept sketches to final assembly

Modeled & fabricated 10+ custom mechanical components in Onshape; iterated geometry to ensure smooth motion

Applied kinematic and energy conservation analysis to optimize track curvature, minimize friction losses, and eliminate derailment

Validated performance through physical testing and redesign cycles

2-axis Pen Plotter

Designed and fabricated a precision 2-axis motion control system from scratch capable of rendering complex 2D geometries

Engineered 20+ custom structural and motion components; optimized frame rigidity and belt tension to improve positional accuracy

Integrated Arduino-based control logic with GRBL firmware to achieve synchronized stepper motor motion

Calibrated motion system to reduce drawing error and improve repeatability

SKILLS AND INTERESTS

Awards: Finalist at International Robotic Invitational, VBC Spring Challenge 1st (0.03%), DECA ICDC 3rd, DECA Business Pitch Challenge 9th, Eagle Scout, FIRST dean's list semi finalist

Technical Skills: Fusion 360, Blender, Onshape, Solidworks, Mechanical Simulation, FDM, P3D Printer, CNC Mill, CO2 Laser Cutter, Welding, Python, Java, MATLAB, Arduino IDE, Raspberry Pi

Interests: DIY Projects, 3D Printing, Investing, Poker, Gym, Reading, Baking